

$$\frac{z+j}{z-j} = w$$

$$\frac{z+j+j}{z-j+j}$$

$$\frac{z+2j}{z} = w$$

$$\frac{z}{z+2j} = w$$

(i) $f(z) = |z|^2$ is not harmonic functions but $f(z) = \ln(|z|^2)$ is harmonic.

Ans: Given,

$$z = |z|^2$$

$$(|z|)^2 = (\sqrt{x^2+y^2})^2$$

$$= x^2 + y^2$$

Diff (i) with respect to x

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2) = 2x + 0 = 2x$$

$$\frac{\partial f}{\partial x} = 2x$$

Again diff (ii) with r.t. to y

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (2x) = 0$$

②

$$\frac{\partial^2 f}{\partial x^2} = 2 \quad \text{--- (i)}$$

New differential: ① w.r.to y

$$f = x^2 + y^2$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x^2 + y^2)$$

$$= 0 + 2y \quad \text{--- (ii)}$$

Again diff (iii) w.r.to x

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (2y)$$

$$\frac{\partial^2 f}{\partial y^2} = 2 \quad \text{--- (iv)}$$

Adding (i) and (iv) we get $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 2 + 2 = 4$

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 2 + 2 = 4$$

which together with (i) and (iv)

thus $f(x, y) = |z|^2$ is not harmonic

for $f(z) = \ln(|z|^2) = \ln(x^2 + y^2)$

diff (v) w.r.to x

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \ln(x^2 + y^2)$$

$$= \frac{1}{x^2 + y^2} \cdot 2x = \frac{2x}{x^2 + y^2}$$

③

$$\text{or, } \frac{g \cdot x}{s_n} = \frac{2n}{n^2 + y^2} \quad \text{--- (vi)}$$

Again,

$$\frac{g}{s_n} \left(\frac{g \cdot x}{s_n} \right) = \frac{g}{s_n} \left(\frac{2n}{n^2 + y^2} \right)$$

$$\Rightarrow \frac{g^2 x}{s_n^2} = \frac{(n^2 + y^2) \cdot \frac{g}{s_n} \cdot 2n}{s_n \cdot \frac{g}{s_n} \cdot (n^2 + y^2)} \quad \text{--- (vii)}$$

$$= \frac{(n^2 + y^2) \cdot 2 - 2n \cdot 2n}{(n^2 + y^2)^2}$$

$$= \frac{2(n^2 + y^2) - 4n^2}{(n^2 + y^2)^2}$$

$$= \frac{2n^2 + 2y^2 - 4n^2}{(n^2 + y^2)^2}$$

$$= \frac{2y^2 - 2n^2}{(n^2 + y^2)^2} \quad \text{--- (viii)}$$

$$= \frac{2(y^2 - n^2)}{(n^2 + y^2)^2} \quad \text{--- (ix)}$$

diff w.r.t to y

$$\frac{g \cdot x}{s_n} = \frac{g}{s_n} (2n(n^2 + y^2))$$

(4) (b)

Hence we will get, $\frac{x^2}{(x^2+y^2)^2} = \frac{2x}{x^2+y^2}$, no

$$\frac{\partial^2 f}{\partial y^2} = \frac{2(x^2-y^2)}{(x^2+y^2)^2} \cdot \frac{1}{x^2+y^2} \quad \text{, simplifying}$$

Adding (i) and (ii) we get,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = \frac{2(x^2-y^2)}{(x^2+y^2)^2} + \frac{2(x^2-y^2)}{(x^2+y^2)^2} = \frac{2(y^2-x^2) + 2(x^2-y^2)}{(x^2+y^2)^2} = \frac{2x^2-2x^2-2y^2+2y^2}{(x^2+y^2)^2} = 0$$

$\Delta f = 0$
(Harmonic)

(v) Check if $f(x,y,z) = x^2+y^2-z^2$ is harmonic

for a function $f(x,y,z)$ to be harmonic

in 3D, $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$

$(x^2+y^2-z^2)$, $\frac{\partial^2}{\partial x^2} = \frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial z^2} = 2$

⑤

Diff $y = x^2 + y^2 - 2xz$ with w.r.t. x

$$\frac{\partial y}{\partial x} = 2x \quad \text{--- (i)}$$

Again diff with x we get

$$\frac{\partial^2 y}{\partial x^2} = 2 \quad \text{--- (ii)}$$

Diff eqn (i) w.r.t. y we get,

$$\frac{\partial y}{\partial y} = 2y \quad \text{--- (iii)}$$

Again,

$$\frac{\partial^2 y}{\partial y^2} = 2 \quad \text{--- (iv)}$$

Diff eqn (i) w.r.t. z we get,

$$\frac{\partial y}{\partial z} = -xz \quad \text{--- (v)}$$

Again,

$$\frac{\partial^2 y}{\partial z^2} = -x \quad \text{--- (vi)}$$

Adding, (ii), (iv) and (vi) we get,

$$\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial y^2} + \frac{\partial^2 y}{\partial z^2} = 2 + 2 - x = 0 \quad \text{(Harmonic)}$$

⑥

⇒ Checking for $\delta(x, y, z) = x^2 + y^2 + z^2$ L.H.D

$$\delta = x^2 + y^2 + z^2 \quad \text{--- (i)}$$

Diff ① w.r.to. x we get L.H.D

$$\frac{\delta \delta}{\delta x} = 2x \quad \text{--- (ii)}$$

Again diff ① w.r.to. y we get

$$\frac{\delta^2 \delta}{\delta x^2} = 2 \quad \text{--- (iii)}$$

Now, Diff ① w.r.to. y

$$\frac{\delta \delta}{\delta y} = 2y \quad \text{--- (iv)}$$

Again diff ① w.r.to. z we get

$$\frac{\delta^2 \delta}{\delta y^2} = 2 \quad \text{--- (v)}$$

Diff ① w.r.to. z we get

$$\frac{\delta \delta}{\delta z} = 2z \quad \text{--- (vi)}$$

Again w.r.to. x we get

$$\frac{\delta^2 \delta}{\delta z^2} = 2 \quad \text{--- (vii)}$$

(7)

Adding (iii), (iv) and (v), we get,

$$\frac{8x^2}{8y^2} + \frac{8x^2}{8y^2} + \frac{8x^2}{8y^2} = 2 - 2 + 2$$

∴ Not Harmonic

∴ $u(x, y) = 3x^2 - 9xy^2$ is Harmonic

(8) Show that $u(x, y) = 3x^2 - 9xy^2$ (i)

Given, $u = 3x^2 - 9xy^2$

Diff (i) w.r.t. to x we get,

$$\frac{\partial u}{\partial x} = 6x - 9y^2$$

Again diff (ii) w.r.t. to x we get,

$$\frac{\partial^2 u}{\partial x^2} = 6$$

Diff (i) w.r.t. to y we get,

$$\frac{\partial u}{\partial y} = -18xy$$

Again diff (ii) w.r.t. to y we get,

$$\frac{\partial^2 u}{\partial y^2} = -18$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 6 - 18 = -12 \neq 0$$

⑧

$$= -18x \quad \text{--- (7)}$$

Adding (6) and (7) we get

$$\frac{82x}{8y^2} + \frac{82x}{8y^2} = 18x - 18x$$

$$= 0$$

(Harmonic) $\frac{1}{x} - \frac{1}{y} = \frac{1}{x} - \frac{1}{y}$ both work (8)

Ans,

iv. Verify $u(x,y) = x^3 - 3xy^2 - 5y$ is Harmonic.

$$\text{Given, } u = x^3 - 3xy^2 - 5y \quad \text{--- (9)}$$

diff (9) w.r.t. to x we get,

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 \quad \text{--- (10)}$$

we get 1/3

Again diff (10) w.r.t. to x we get 1/3

$$\frac{\partial^2 u}{\partial x^2} = 6x \quad \text{--- (11)}$$

Now, diff (9) w.r.t. to y we get

$$\frac{\partial u}{\partial y} = -6xy - 5 \quad \text{--- (12)}$$

$$= -3(2y + 0) - 5$$

$$= -6xy - 5 \text{ --- (iv)}$$

Again diff (iv) w.r.t. to y we get,

$$\begin{aligned} \frac{\partial^2 u}{\partial y^2} &= -6 \left(x \frac{\partial}{\partial y} y + y \frac{\partial}{\partial y} x \right) \\ &= -6x \text{ --- (v)} \end{aligned}$$

Now, Adding (iv) and (v) we get,

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 6x - 6x \\ &= 0 \end{aligned}$$

Thus $u(x, y)$ is Harmonic. Ans.

$$(b) u = e^y \sin x$$

Given $u = e^{-y} \sin x$ --- (1) [L.H.S.]

Diff (1) w.r.t. to x we get,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \sin x \frac{\partial}{\partial x} (e^{-y}) + e^{-y} \frac{\partial}{\partial x} \sin x \\ &= \sin x \cdot 0 + e^{-y} \cos x \\ &= e^{-y} \cos x \text{ --- (2)} \end{aligned}$$

10

Again, diff (iii) w.r.t. to x gives -

$$\frac{\partial^2 u}{\partial x^2} = e^{-y} - \sin x \cdot \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = e^{-y} - \sin x \cdot \frac{\partial}{\partial x} (e^{-y} \sin x) = e^{-y} - \sin x \cdot e^{-y} \cos x = e^{-y} (1 - \sin^2 x) = e^{-y} \cos^2 x$$

$$= e^{-y} \cos^2 x$$

Now, Diff (iv) w.r.t. to y we get

$$\frac{\partial u}{\partial y} = \sin x \cdot e^{-y} \cdot (-1) = -\sin x \cdot e^{-y}$$

$$= -e^{-y} \sin x$$

Again diff (iv) w.r.t. to y we get, $\frac{\partial^2 u}{\partial y^2} = -e^{-y} \sin x$

$$\frac{\partial^2 u}{\partial y^2} = -\sin x \cdot e^{-y} \cdot (-1) = \sin x \cdot e^{-y}$$

Adding (iii) and (v) we get,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^{-y} \cos^2 x - \sin x \cdot e^{-y} \cos x + \sin x \cdot e^{-y} \cos x = e^{-y} \cos^2 x$$

$$= e^{-y} \cos^2 x$$

(Harmonics)

$$\text{Ans } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^{-y} \cos^2 x$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^{-y} \cos^2 x$$

(11)

(c) no is same

(d) is also same, we can do it

by previous methods mention above

$$(e) u = e^n (n \cos y - y \sin y) \quad \text{--- (1)}$$

Given, $u = e^n (n \cos y - y \sin y)$

Diff w.r.to n we get,

$$\frac{\partial u}{\partial n} = \frac{\partial}{\partial n} (e^n (n \cos y - y \sin y)) + (n \cos y - y \sin y) + (n \cos y - y \sin y) \frac{\partial e^n}{\partial n}$$

$$= e^n (n \cos y - y \sin y) + (n \cos y - y \sin y) + e^n \cos y$$

$$= e^n (n \cos y - y \sin y) + (n \cos y - y \sin y) + e^n \cos y$$

$$= e^n (n \cos y - y \sin y) + (n \cos y - y \sin y) + e^n \cos y$$

we get,

Again diff w.r.to n we get,

$$\frac{\partial^2 u}{\partial n^2} = \frac{\partial}{\partial n} (e^n (n \cos y - y \sin y) + (n \cos y - y \sin y) + e^n \cos y)$$

$$= e^n (n \cos y - y \sin y) + (n \cos y - y \sin y) + e^n \cos y$$

$$= e^n (n \cos y - y \sin y) + (n \cos y - y \sin y) + e^n \cos y + e^n \cos y$$

(10)

$$\textcircled{5} \quad = e^n (n \cos y - y \sin y) + 2e^n \cos y$$

Diff ① w.r.to. y

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y} e^n (n \cos y - y \sin y) \\ &= e^n \left(\frac{\partial}{\partial y} (n \cos y - y \sin y) \right) \\ &= e^n (n(-\sin y) - (y \cos y + \sin y)) + (n \cos y - y \sin y) \frac{\partial}{\partial y} e^n \\ &= e^n (-n \sin y - y \cos y - \sin y) + (n \cos y - y \sin y) \frac{\partial}{\partial y} e^n \end{aligned}$$

Again diff ② w.r.to. y we get,

$$\begin{aligned} \frac{\partial^2 u}{\partial y^2} &= e^n \left(\frac{\partial}{\partial y} (-n \sin y - y \cos y - \sin y) + (n \cos y - y \sin y) \frac{\partial}{\partial y} \right) \\ &= e^n \left(-n \cos y - (y \sin y + \cos y) + (n \cos y - y \sin y) \right) \\ &= e^n (-n \cos y - y \sin y + \cos y - \cos y + y \sin y) \\ &= e^n (-n \cos y + y \sin y - \cos y) \end{aligned}$$

Adding ③ and ④ we get,

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= e^n (n \cos y - y \sin y) + 2e^n \cos y \\ &+ e^n (-n \cos y + y \sin y - \cos y) \\ &= e^n (n \cos y - y \sin y + 2 \cos y - n \cos y + y \sin y - \cos y) \\ &= e^n \cos y \end{aligned}$$

(13)

$$= e^{n \cos y} \quad (70)$$

(Not Harmonic)

And the rest of the maths

are same (111) Task
Home $p - u_0 \rightarrow 1$

$$|z| = \sqrt{x^2 + y^2}$$

$$|z| = 2$$